COMP47750
Machine Learning with Python

Autumn 2026 Assignment 1
Evaluating Categorical Encoding Methods

Overview

This assignment explores different categorical encoding strategies for use with Naive Bayes classifiers. You will compare three encoders commonly used for transforming categorical variables and evaluate their behaviour under different training/testing setups.
The dataset to be used in this assignment is provided in the csv file diabetes_70k.csv.

Task 1 — OneHotEncoder vs OrdinalEncoder with Naive Bayes

Using scikit-learn, evaluate the following encoder–classifier combinations:

- OneHotEncoder → MultinomialNB
- OrdinalEncoder → CategoricalNB

For this task:

1. Apply each encoder to the dataset
2. Use a 50:50 train–test split.
3. Train the classifier.
4. Repeat the experiment multiple times (e.g., 20 runs) using different random seeds.
5. Report the average accuracy for each encoder.

random seeds for the test-train split

Task 2 — CatBoostEncoder with GaussianNB

Use the CatBoostEncoder from the category_encoders package:

```
pip install category_encoders
```

change int to str to use catboost!

Because CatBoostEncoder produces continuous-valued features, use:

- CatBoostEncoder → GaussianNB

Repeat the same evaluation procedure as in Task 1.

Task 3 — Fit Encoders *After* Splitting the Data

Repeat Tasks 1 and 2, but change the workflow:

1. Split the dataset into training and testing sets first.
2. Fit each encoder only on the training data.
3. Transform both training and test sets using the fitted encoder.
4. Evaluate the same three encoder–classifier combinations.

Explain data leakage and how it relates to this task.

Task 4 — Visual Comparison of Results

Create a bar chart summarising the average accuracy for:

- OneHotEncoder + Naive Bayes
- OrdinalEncoder + Naive Bayes
- CatBoostEncoder + GaussianNB

Include results from:

- Encoding before splitting
- Encoding after splitting

Your plot should show how performance varies across encoders and evaluation setups.

Task 5 — Markdown Commentary

In your notebook, write a short analysis discussing:

- Which encoder performed best overall
- How the choice of Naive Bayes variant affected results
- The effect of fitting encoders before vs. after splitting
- Any practical recommendations for handling categorical data in ML pipelines

Submission Requirements

Submit the following through Brightspace:

1. Your Jupyter notebook (.ipynb) BMI is gaussian so should do best
2. An HTML export of the notebook
3. If your notebook takes time to run, do not clear outputs
4. This is an individual (not group!) assignment and genAI usage is **not** permitted